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The germline mutational process in rhesus macaque and its implications for phylogenetic dating

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Abstract

Background: Understanding the rate and pattern of germline mutations is of fundamental importance for understanding evolutionary processes. **Results:** Here we analyzed 19 parent-offspring trios of rhesus macaques (*Macaca mulatta*) at high sequencing coverage of ∼76× per individual and estimated a mean rate of 0.77×10^{-8} *de novo* mutations per site per generation (95% CI: 0.69×10^{-8} to 0.85×10^{-8}). By phasing 50% of the mutations to parental origins, we found that the mutation rate is positively correlated with the paternal age. The paternal lineage contributed a mean of 81% of the *de novo* mutations, with a trend of an increasing male contribution for older fathers. Approximately 3.5% of *de novo* mutations were shared between siblings, with no parental bias, suggesting that they arose from early development (postzygotic) stages. Finally, the divergence times between closely related primates calculated on the basis of the yearly mutation rate of rhesus macaque generally reconcile with divergence estimated with molecular clock methods, except for the Cercopithecoidea/Hominoidea molecular divergence dated at 58 Mya using our new estimate of the yearly mutation rate.

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